



Mirjabbar Badalov / ID: 285269

MPA-MEM 25/26L

Semester Project

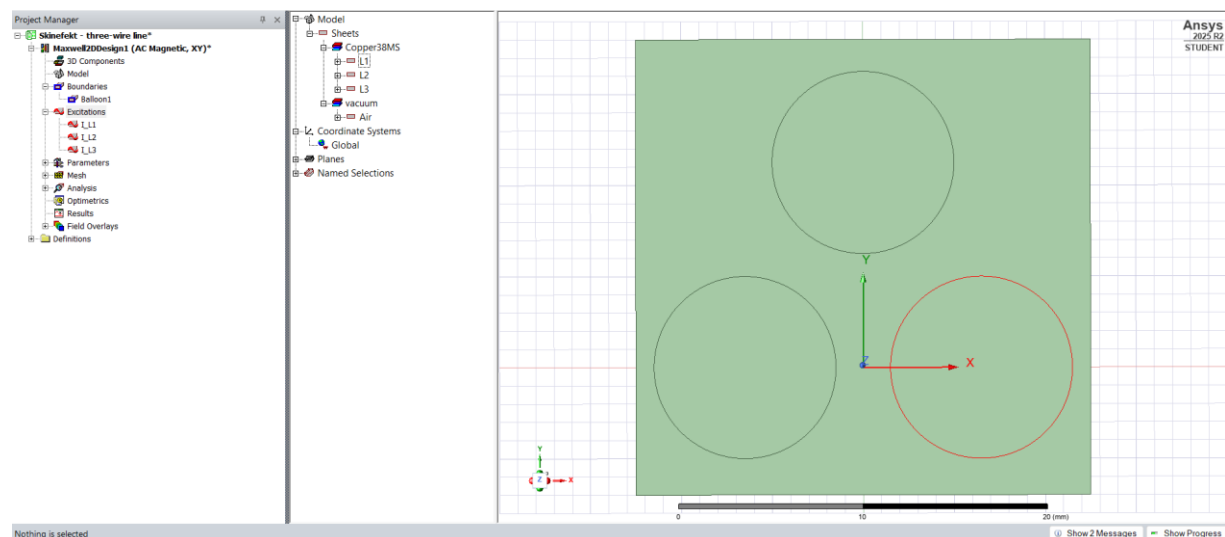
Skinefekt - three-wire line

An alternating current $I_{\max} = 150$ A flows through the conductors of circular cross-section arranged as shown. Consider the same direction of currents in all conductors and the currents $I_{L1\max} = 150$ A, $I_{L2\max} = 150 \angle -120^\circ$ and $I_{L3\max} = 150 \angle 120^\circ$ A. The diameter of the conductors is 10 mm, the location of the conductors is shown in the figure. The specific electrical conductivity of the conductors is $\sigma = 38$ MS/m. For frequencies 1 Hz - 10 kHz.

1. Introduction

The objective of this project is to simulate and analyze the electromagnetic behavior of a three-wire alternating current (AC) system using Ansys Maxwell 2D. The system consists of three parallel conductors with circular cross-sections (10 mm diameter) and a specific electrical conductivity ($\sigma = 38$ MS/m). The conductors carry an alternating current with a maximum amplitude of 150 A, featuring symmetrical phase shifts (0° , -120° , 120°).

The primary focus of this simulation is to observe and quantify the skin effect and proximity effect across a frequency sweep ranging from 1 Hz to 10 kHz. Using the AC Magnetic solver, various electromagnetic parameters were calculated, including magnetic flux lines, current density distribution, magnetic energy density, and Joule losses. Additionally, derived quantities such as the effective AC resistance, inductance, and the electromagnetic forces acting on the conductors were extracted and compared against theoretical expectations to validate the simulation model.

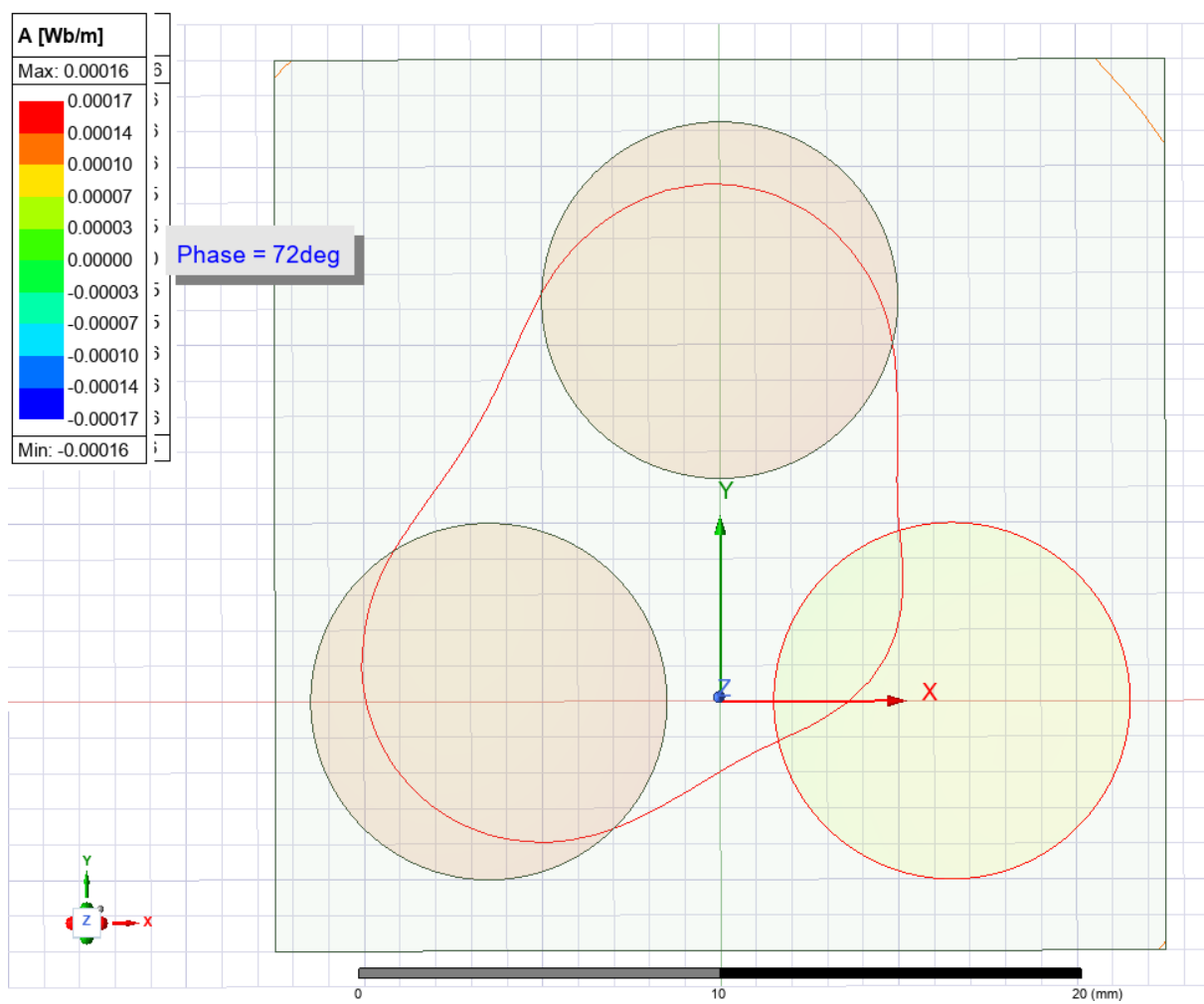
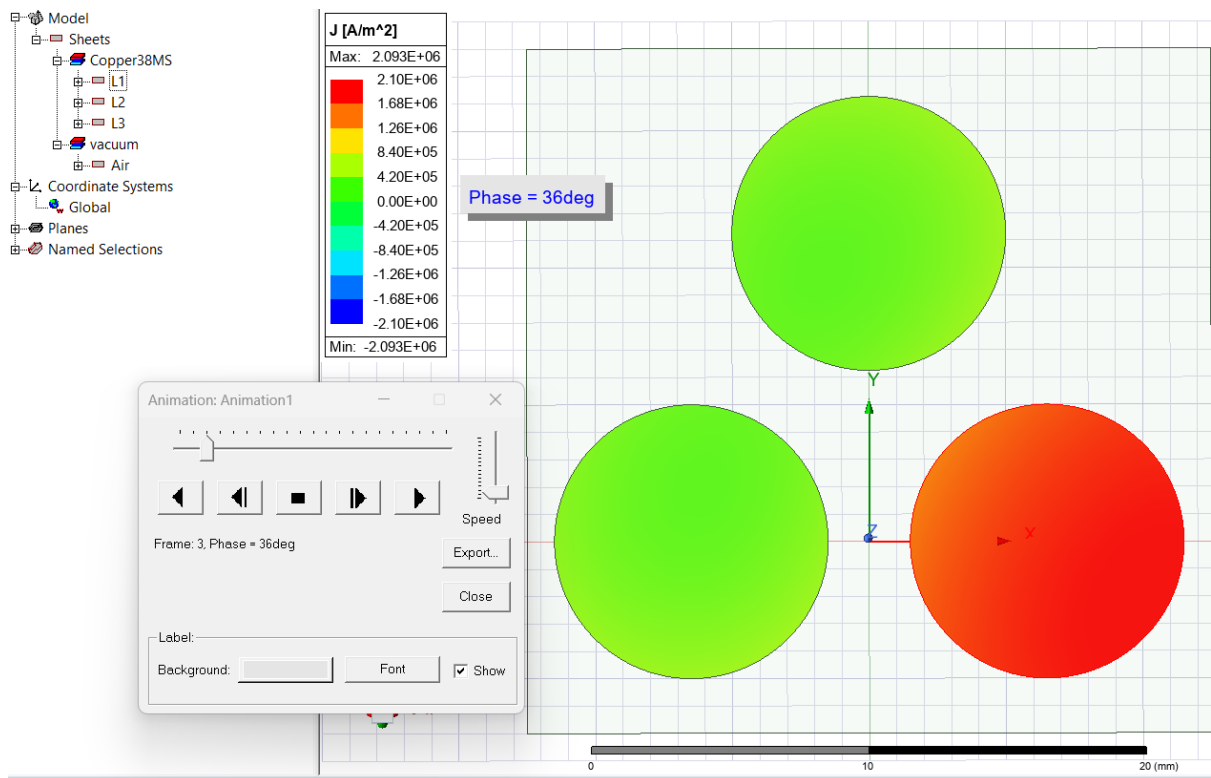


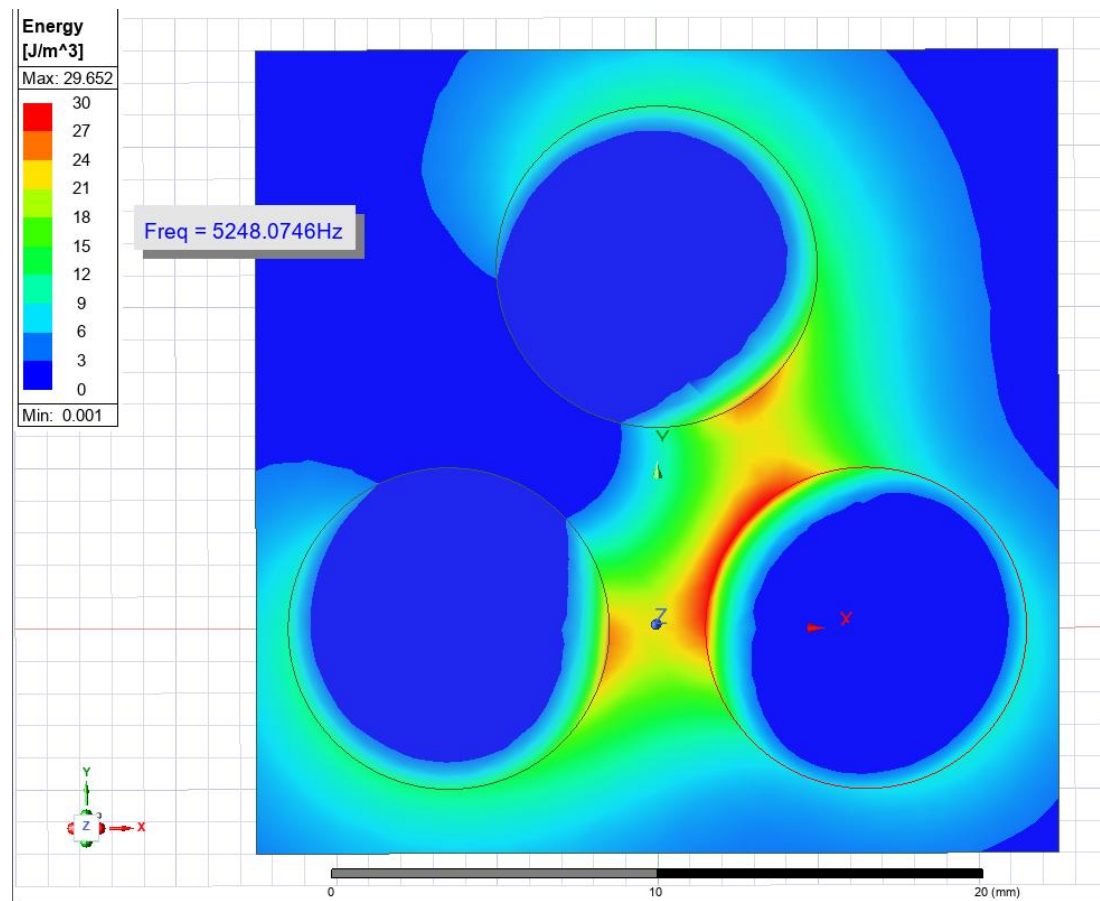
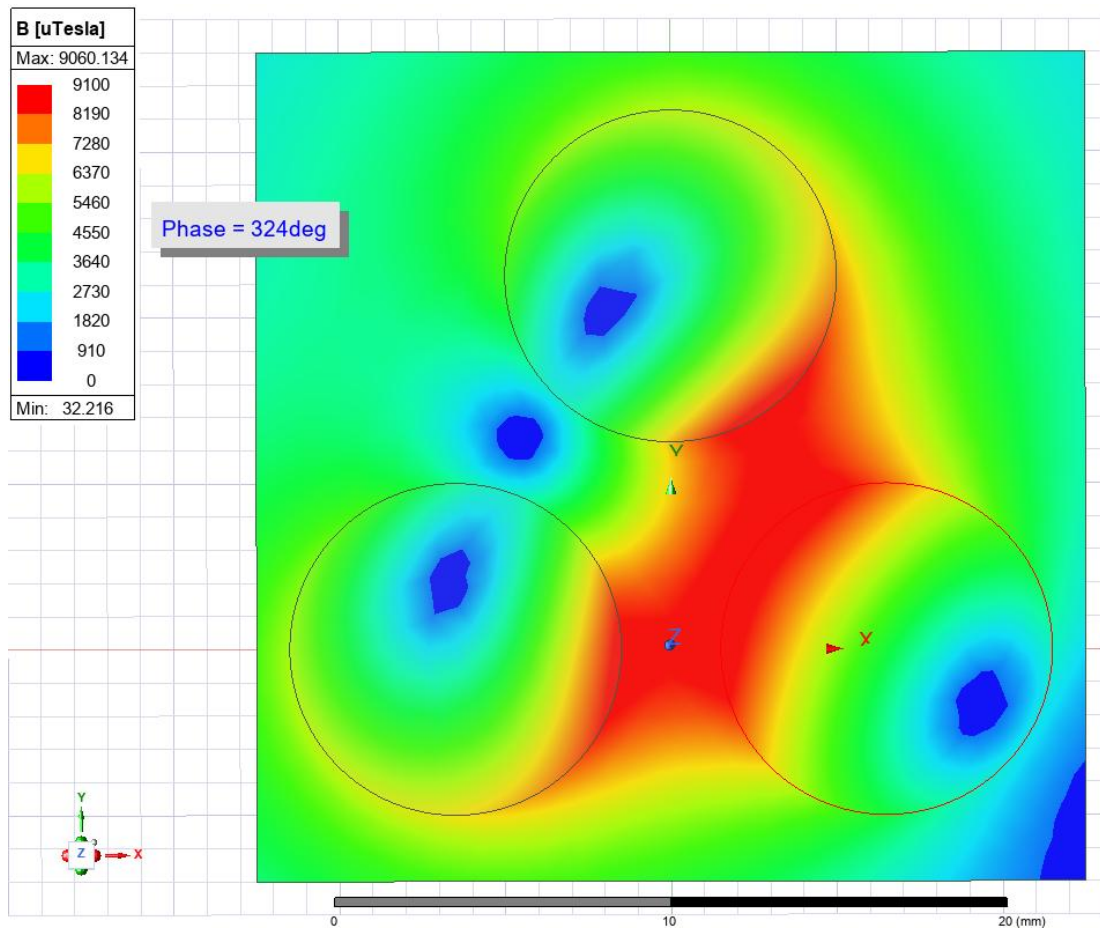
2. Difficulties Encountered and Troubleshooting

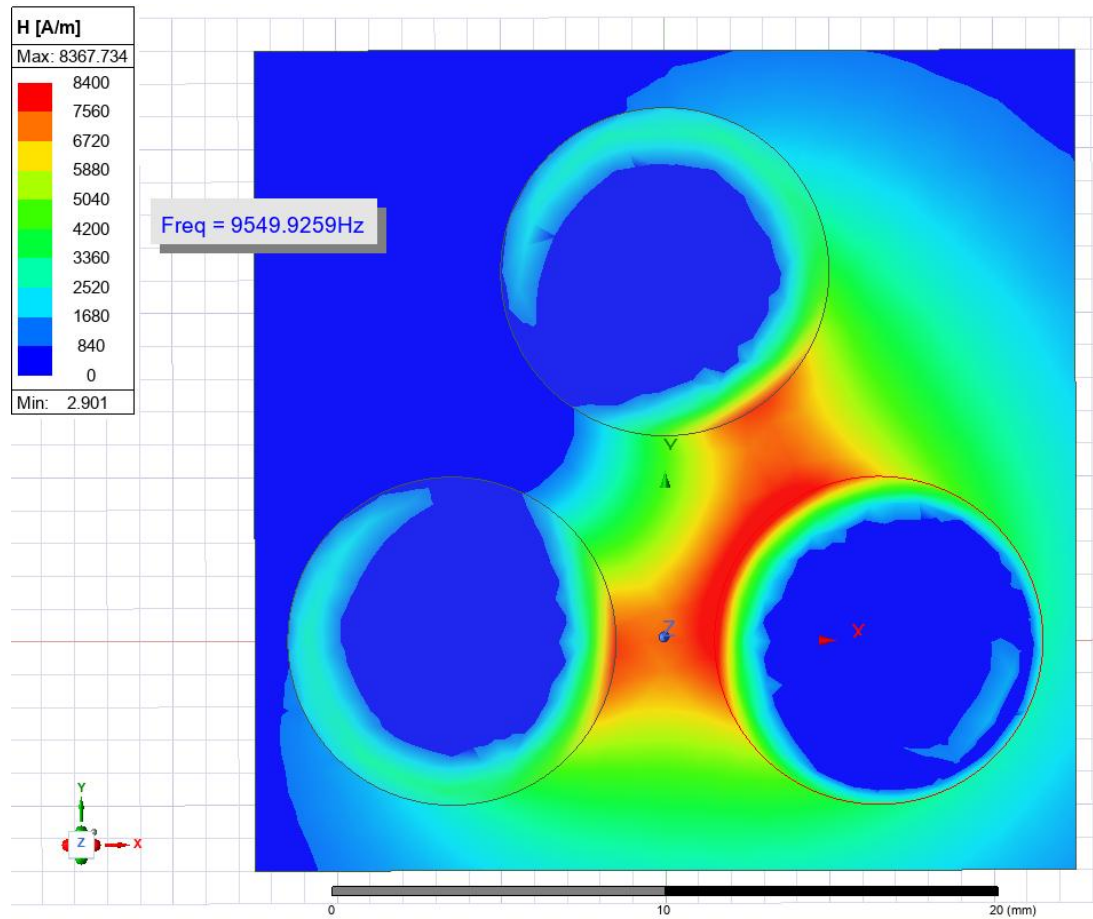
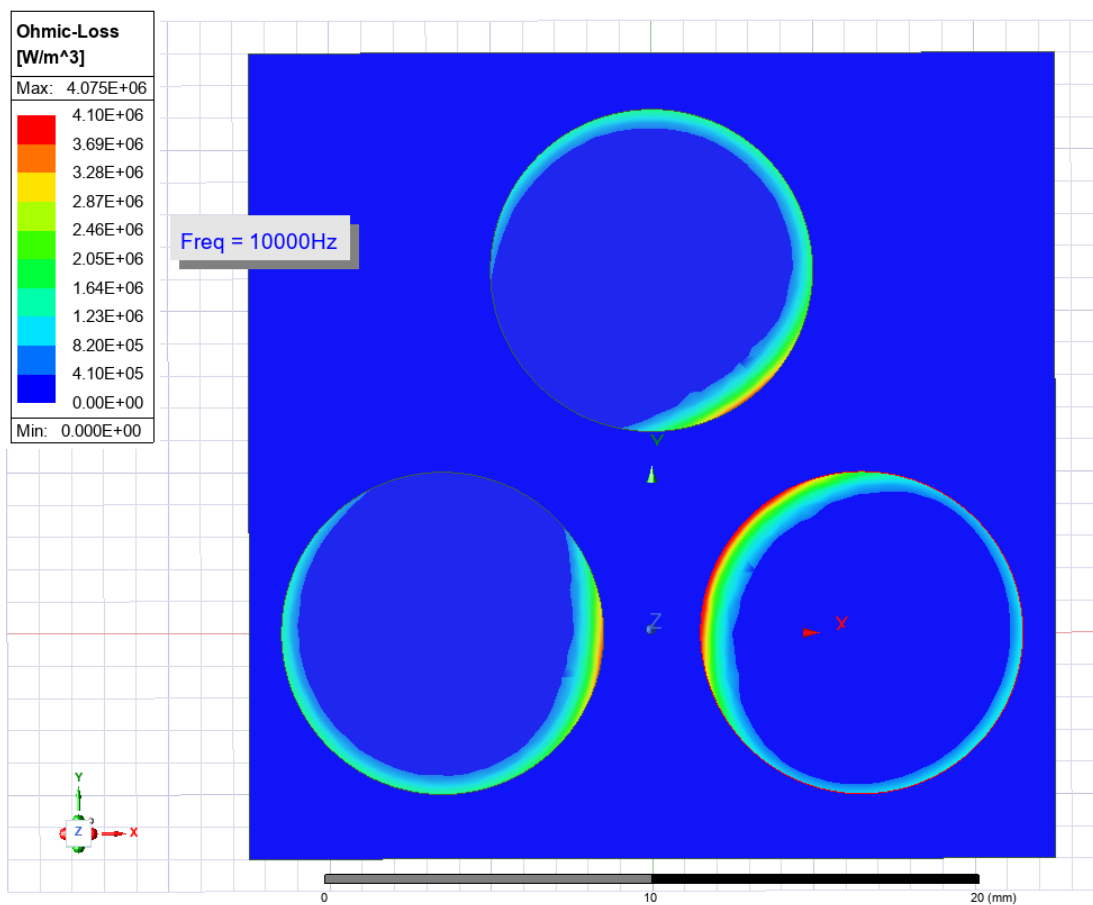
While the geometric setup and boundary conditions were established successfully, a significant challenge arose during the data extraction and reporting phase of the project. Specifically, the simulation ran without errors (yielding a successful calculation), but the required quantities for Resistance, Inductance, and Force failed to appear when attempting to generate rectangular plots and data tables.

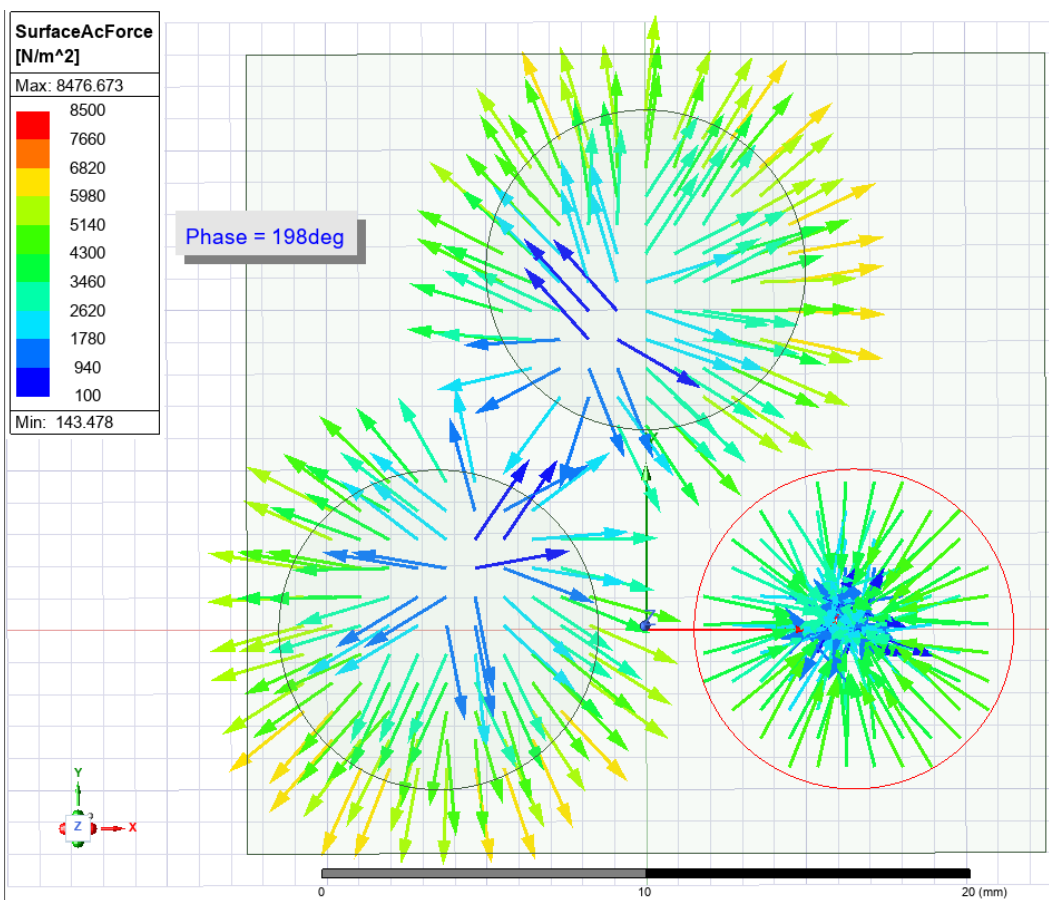
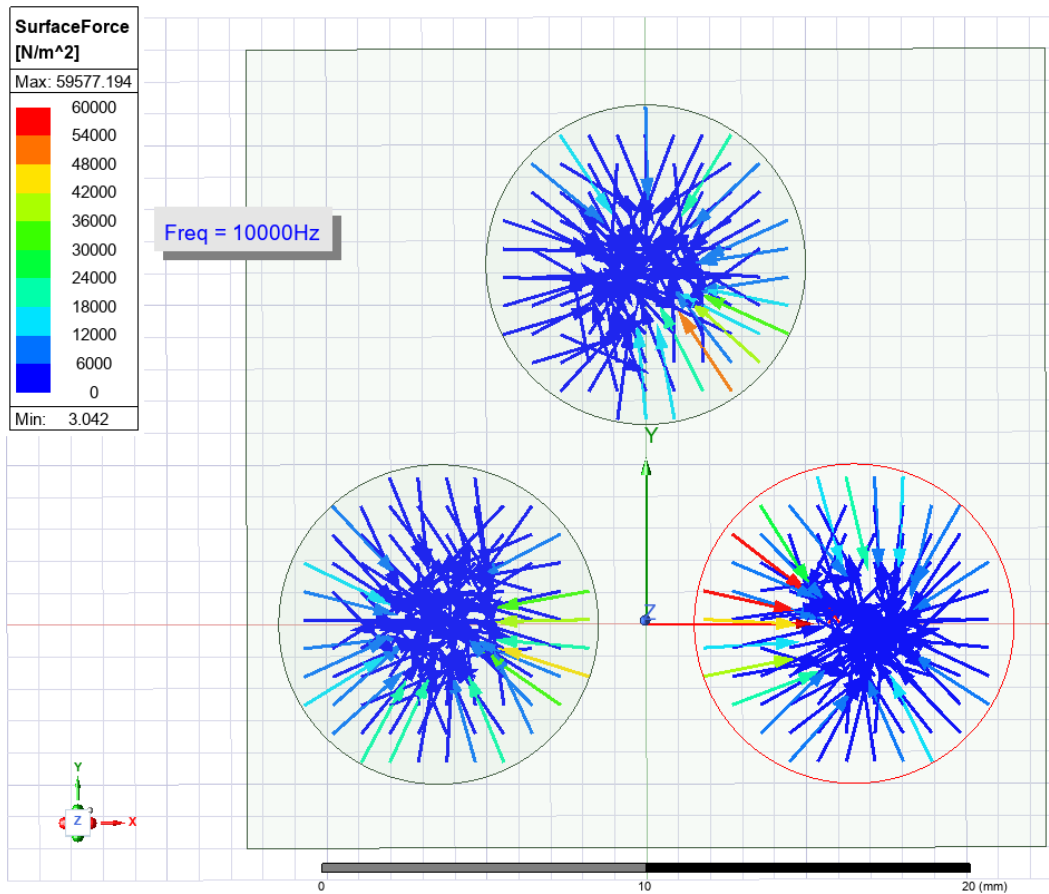
Through systematic troubleshooting, the following solver mechanics in Ansys Maxwell were identified as the root cause and subsequently resolved:

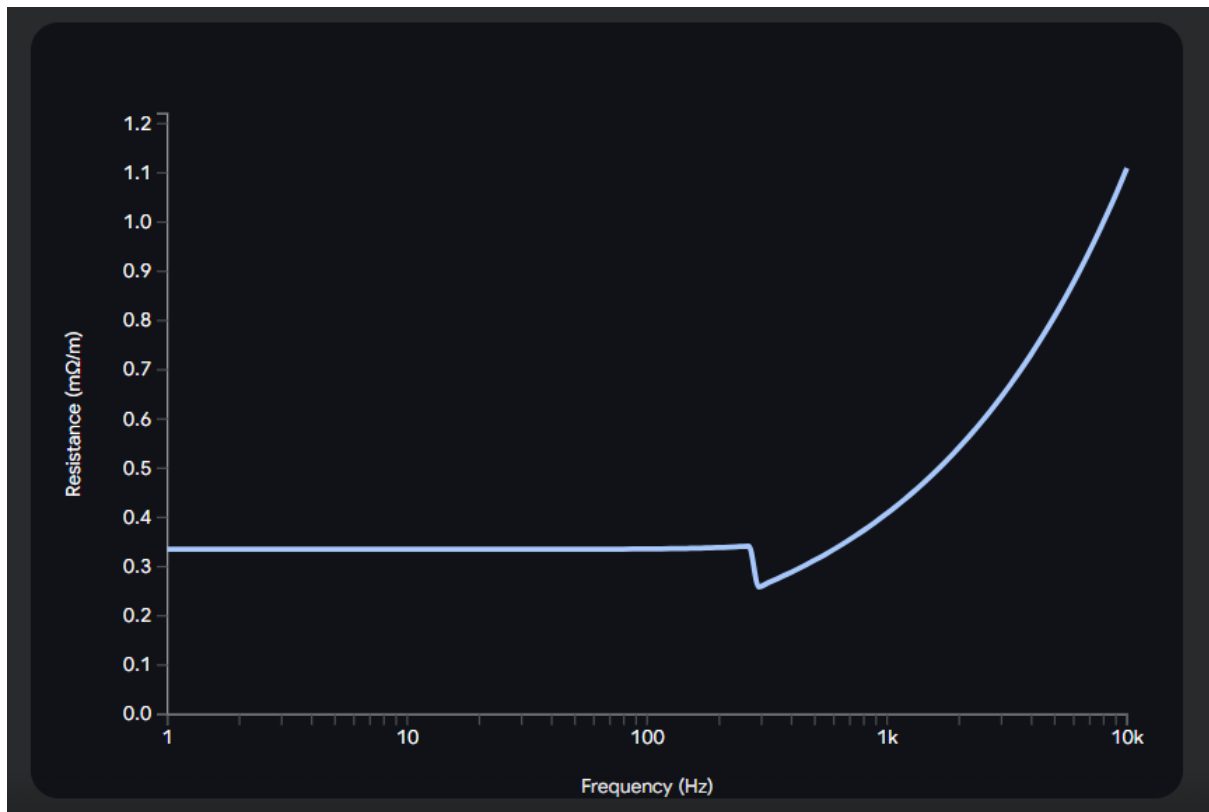
- **Explicit Parameter Assignment:** By default, the Ansys AC Magnetic solver calculates foundational field data (such as Magnetic Flux Density, B, and Current Density, J) to save computational resources. It was discovered that derived values—specifically the Resistance/Inductance matrices and Force calculations—must be explicitly assigned to the conductors in the Parameters menu *prior* to running the simulation. Without this manual assignment, the "Matrix" and "Force" categories remain empty in the Results menu.
- **Frequency Sweep Data vs. Mesh Data:** A secondary issue occurred where rectangular plots for AC Resistance versus Frequency generated entirely blank graphs. This was traced to the report's Context settings. By default, Ansys attempts to plot using the Setup1 : LastAdaptive solution, which only contains data for the single frequency used to generate the finite element mesh.
- **Resolution:** To resolve this, a full Frequency Sweep (1 Hz to 10 kHz) was added to the solution setup. Crucially, the model had to be re-analyzed specifically for the sweep *after* assigning the Matrix parameters. Once the calculation finished, the report context was manually changed from LastAdaptive to Sweep, which successfully populated the frequency-dependent curves.











Could not get values shown graph from the results tab. That's why uploaded graph with values of my calculation.

Some Screenshots from my project properties:

Solve Setup

General | Convergence | Expression Cache | Solver | Frequency Sweep | Defaults

Name: ☒ Enabled

Adaptive Setup

Maximum Number of Passes:

Percent Error:

Parameters

☒ Solve Fields Only

Solve Matrix: ☒ After last pass ☐ Only after converging

Solve Setup

General | Convergence | Expression Cache | Solver | Frequency Sweep | Defaults

Adaptive Frequency: Hz

Nonlinear Residual:

☐ Smooth BH Curve

☐ Number of Nonlinear Iterations

Minimum:

Maximum:

Thermal Two-way Coupling

Relaxation

Note: This factor is applicable only when Enable Feedback is checked in Temperature of Object dialog. The range for this factor is [0,1].

☐ Import mesh

Solve Setup

General | Convergence | Expression Cache | Solver | Frequency Sweep | Defaults

Frequency Sweeps [201 points defined]

	Distribution	Start	End		
1	Log Scale	1Hz	10000Hz	Samples	50

Select Definition

Materials | Material Filters

Search Parameters

Search by Name

Search Criteria

☒ by Name ☐ by Property

Libraries ☒ Show Project definitions ☐ Select

[sys] Materials

	Name	Location	Origin	Relative Permittivity	Relative Permeability	Bulk Conductivity
	Copper38MS	Project		1	1	38000000

